



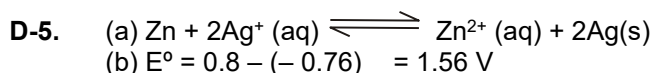
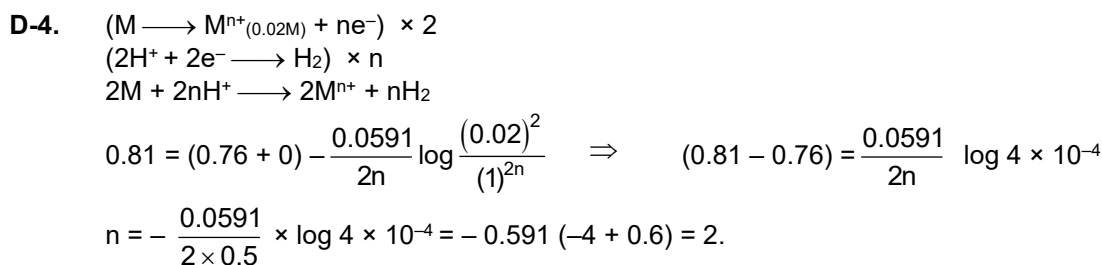
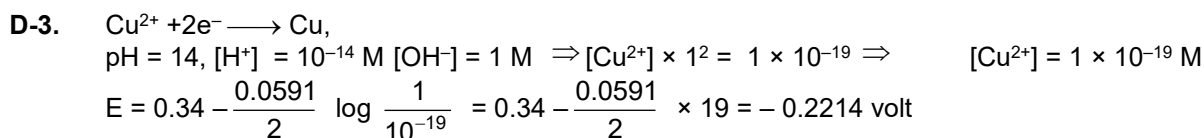
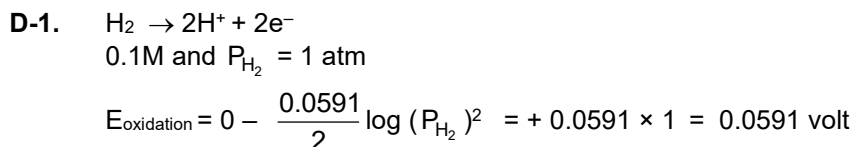
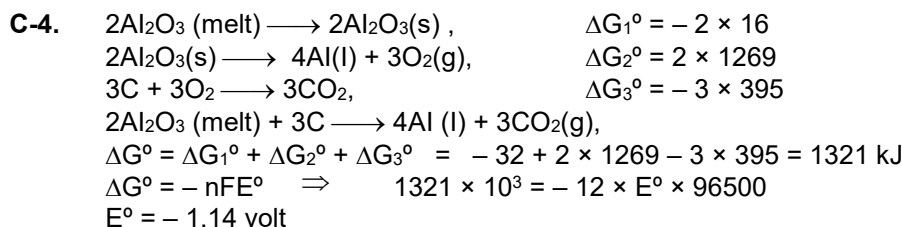
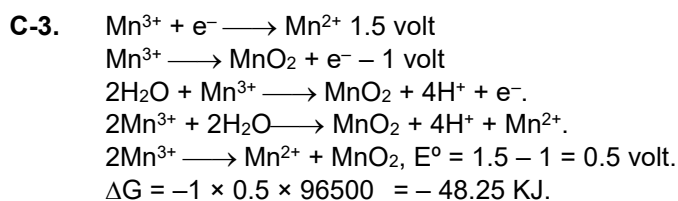
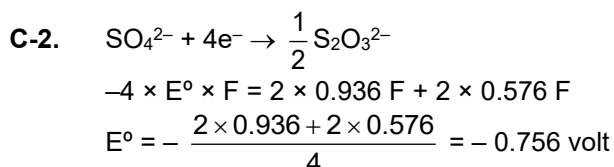
SOLUTION OF ELECTROCHEMISTRY

EXERCISE # 1

PART - I

B-1. Lowest standard reduction potential highest reducing power.

B-2. Lowest standard reduction potential highest reducing power.

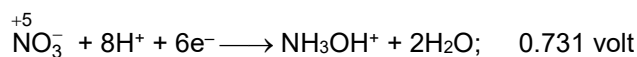
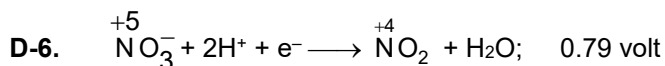




$$(c) \quad 1.6 = 1.56 - \frac{0.059}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2} \quad \text{or} \quad 1.356 = \log \frac{[\text{Ag}^+]^2}{[\text{Zn}^{2+}]}$$

$$\text{or} \quad \log \frac{[0.1]^2}{[\text{Zn}^{2+}]} = 1.356 \quad \text{or} \quad \frac{0.01}{[\text{Zn}^{2+}]} = 22.7$$

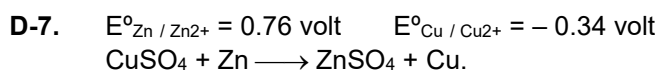
$$\text{or} \quad [\text{Zn}^{2+}] = \frac{0.01}{22.7} = 4.4 \times 10^{-4} \text{ M}$$



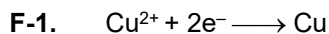
$$E = 0.79 - \frac{0.0591}{1} \log \frac{1}{[\text{H}^+]^2} = 0.731 - \frac{0.0591}{6} \log \frac{1}{[\text{H}^+]^8}$$

$$0.79 + 2 \times \frac{0.0591}{2} \log [\text{H}^+] = 0.731 + \frac{0.0591}{6} \times 8 \times \log [\text{H}^+]$$

$$0.059 = 0.059 \left(2 - \frac{8}{6}\right) \text{ pH} \quad \Rightarrow \text{pH} = \frac{6}{4} = 1.5$$



$$0 = (0.76 + 0.34) - \frac{0.0591}{2} \log \left(\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \right) \quad \Rightarrow \quad \log \left(\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \right) = 37.22$$



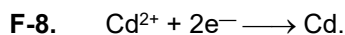
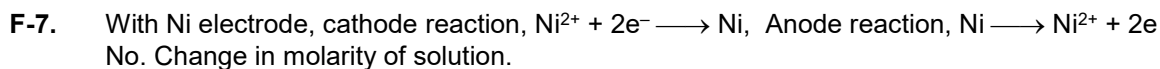
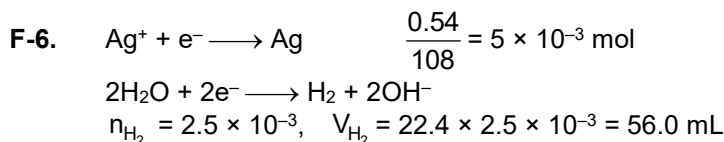
$$\text{No. of Faraday required} = 2 \text{ F} = 2 \times 6.023 \times 10^{23} = 12.04 \times 10^{23}$$

F-2. $0.108 = \frac{E}{96500} \times 0.5 \times 193 \quad E = 108 \text{ g/eq.}$

F-3. $\frac{W_{\text{Ag}}}{W_{\text{metal}}} = \frac{E_{\text{Ag}}}{E_{\text{M}}} = \frac{108}{\frac{108}{n}} = n \quad n = \frac{0.5094}{0.2653} = 1.92 \quad n = 2.$

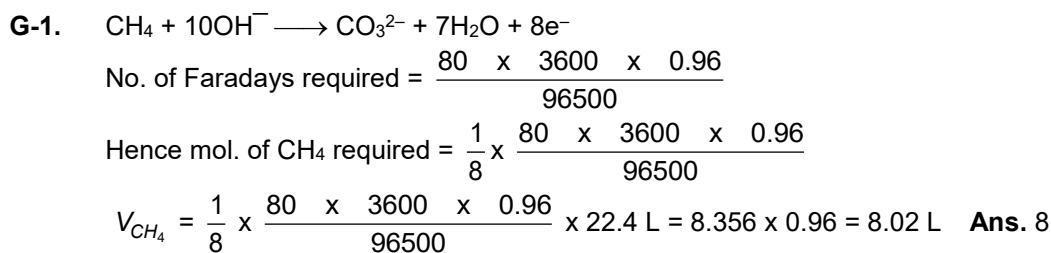
F-4. $\frac{2.977}{106.4/n} = \frac{3 \times 1 \times 60 \times 60}{96500} \quad \Rightarrow \quad n = 4.$

F-5. $10.8 \times 80 \times 5 \times 10^{-4} = \frac{108}{1 \times 96500} \times 2 \times t \quad \Rightarrow \quad t = 193 \text{ sec}$



$$\text{Cd required for 2g of Hg} = \frac{12 \times 2}{88} = 0.2727 \text{ g.}$$

$$t = \frac{2 \times 96500 \times 0.2727}{112.4 \times 5} = 93.65 \text{ Sec.}$$



H-1. $K = \frac{1}{245} \times \frac{4}{7} = 2.332 \times 10^{-3} \text{ mho cm}^{-1}$.
 $\Lambda_m = \frac{2.33 \times 10^{-3} \times 1000}{0.1} = 23.32 \text{ mho cm}^2 \text{ mol}^{-1}$.

H-2. $\Lambda_{\text{eq}} = 97.1 \text{ Scm}^2 \text{ eq}^{-1}$, $C = 0.1 \text{ N}$
 $A = 1.5 \text{ cm}^2$, $l = 0.5 \text{ cm}$
 $\Lambda_{\text{eq}} = \frac{1000 \times \left(\frac{1}{R} \times \frac{l}{A}\right)}{C} \Rightarrow 97.1 = \frac{1000}{0.1} \times \frac{1}{R} \times \frac{0.5}{1.5}$
 $R = 34.33 \Omega \Rightarrow i = \frac{V}{R} = \frac{5}{34.33} = 0.1456 \text{ amp}$

H-3. $C = 0.1 \text{ N}$, $K = 1.12 \times 10^{-2} \text{ Scm}^{-1}$, $R = 65 \Omega$
 $K = \frac{1}{R} \times \frac{l}{A}$ $\frac{l}{A} = \text{cell constant} = 1.12 \times 10^{-2} \times 65 = 0.728 \text{ cm}^{-1}$

I-1. $\Lambda_m^\circ(\text{NH}_4\text{Cl}) = 150 = \Lambda_m^\circ(\text{NH}_4^+) + \Lambda_m^\circ(\text{Cl}^-)$
 $\Lambda_{\text{OH}^-}^\circ = 198$, $\Lambda_{\text{Cl}^-}^\circ = 76$, $\Lambda_m(\text{NH}_4\text{OH}) = 9.6$, $C = 0.01$
 $\Lambda_m^\circ(\text{NH}_4\text{OH}) = 150 - 76 + 198 = 272$
 $\alpha = \frac{\Lambda_m(\text{NH}_4\text{OH})}{\Lambda_m^\circ(\text{NH}_4\text{OH})} = \frac{9.6}{272} = 0.0353$

I-2. $\Lambda_m(\text{NaA}) = 83 \text{ S cm}^2 \text{ mol}^{-1}$ $\Lambda_m(\text{NaCl}) = 127 \text{ S cm}^2 \text{ mol}^{-1}$ $\Lambda_m(\text{HCl}) = 426 \text{ S cm}^2 \text{ mol}^{-1}$
 $\Lambda_m(\text{HA}) = 83 + 426 - 127 = 382 \text{ S cm}^2 \text{ mol}^{-1}$.

I-3. $\alpha = \frac{7.36}{390.7}$ $K_a = C\alpha^2 = 0.05 \times \left(\frac{7.36}{390.7}\right)^2 = 1.77 \times 10^{-5} \text{ mol/lit}$

I-4. $K_{\text{AgCl}} = 2.28 \times 10^{-6} \text{ Scm}^{-1}$, $138.3 = \frac{1000 \times 2.28 \times 10^{-6}}{S}$
 $S = 1.65 \times 10^{-5}$ and $K_{\text{sp}} = (S)^2 = 2.72 \times 10^{-10} \text{ M}^2$.

PART - II

- A-1.** In galvanic cell/electro chemical cell electrical energy is produced due to some chemical reaction.
A-2. Salt bridge complete the electrical circuit and minimises the liquid-liquid junction potential.
A-3. Agar-Agar is a gelatin, it used in salt bridge along with KCl electrolyte.



A-4. $E_{\text{cell}} = E^{\circ}_{\text{Ni} / \text{Ni}^{2+}} + E^{\circ}_{\text{Ag}^{+} / \text{Ag}}$
 $= 0.25 + 0.80 = \mathbf{1.05 \text{ Volt.}}$

B-1. E° is intensive property and it does not depend on mass of F_2 taking part.

B-2. Lowest S.R.P., highest reducing power.

B-3. $E^{\circ}_{\text{Cu}^{2+} / \text{Cu}} = 0.34$ $E^{\circ}_{\text{Fe}^{2+} / \text{Fe}} = -0.44 \text{ volt}$
 So Cu can't displace Fe^{2+} .

B-4. Cu can't displace Al^{3+} ion from aluminium nitrate.

B-5. Lower S.R.P. containing ion can displace higher S.R.P. containing ion.

B-6. Lowest S.R.P., highest reducing power.

B-7. KCl can make precipitate with AgNO_3 , $\text{Pb}(\text{NO}_3)_2$ so can't be used along these electrolyte.

C-1. $\text{Fe}^{3+} + 3\text{e}^{-} \longrightarrow \text{Fe}$, -0.036 volt
 $\text{Fe} \longrightarrow \text{Fe}^{2+} + 2\text{e}^{-}$ 0.44 volt
 $\text{Fe}^{3+} + \text{e}^{-} \longrightarrow \text{Fe}^{2+}$
 $+ 3 \times 0.036 - 2 \times 0.44 \times f = -1 \times E^{\circ} \times f$
 $E^{\circ} = \mathbf{0.772 \text{ Volt}}$

C-2. $\text{Cu}^{+} + \text{e}^{-} \longrightarrow \text{Cu}$, $E^{\circ} = x_1 \text{ Volt}$
 $\text{Cu}^{2+} + 2\text{e}^{-} \longrightarrow \text{Cu}$, $x_2 \text{ Volt}$
 $\text{Cu} \longrightarrow \text{Cu}^{+} + \text{e}^{-}$ $-x_1 \text{ Volt}$
 $\text{Cu}^{2+} + \text{e}^{-} \longrightarrow \text{Cu}^{+}$
 $-2 \times x_2 \times f + 1 \times x_1 \times f = -1 \times E^{\circ} \times f$
 $E^{\circ} = \mathbf{2x_2 - x_1}$

C-3. For spontaneous reaction in every condition
 $E_{\text{cell}} > 0$, $\Delta G < 0$ and Q (reaction quotient) $< K$ (equilibrium constant).

D-1. $E = 1.1 - \frac{0.0591}{2} \log \frac{0.1}{0.1} \Rightarrow \mathbf{E = 1.10 \text{ Volt}}$

D-2. $\text{H}_2(\text{Pt}) (1 \text{ atm}) | \text{H}_3\text{O}^{+} || \text{Ag}^{+}(\text{xM}) | \text{Ag}$

$$1.0 = (0 + 0.8) - \frac{0.06}{1} \log \frac{[\text{H}^{+}]}{x}$$

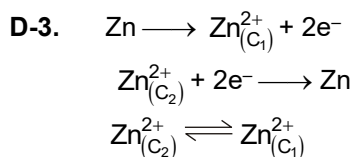
$$-\frac{0.2}{0.06} = \log \frac{[\text{H}^{+}]}{x}$$

$$\frac{10}{3} = \text{pH} + \log x$$

$$\log x = -1.7$$

$$\frac{10^{-5.5}}{x} = 1.62 \times 10^{-4}$$

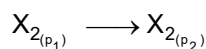
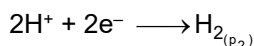
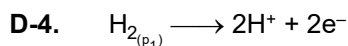
$$\mathbf{x = 2 \times 10^{-2} \text{ M}}$$





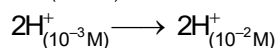
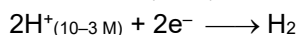
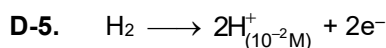
$$E = 0 - \frac{0.0591}{2} \log \frac{C_1}{C_2}$$

$E \rightarrow +ve$ When $C_1 < C_2$



$$E = 0 - \frac{0.0591}{2} \log \frac{p_2}{p_1}$$

$P_2 < P_1$ for $E \rightarrow +ve$



$$E = 0 - \frac{0.0591}{2} \log \left(\frac{10^{-1}}{10^{-2}} \right)^2$$

$E \rightarrow -ve$ (Non spontaneous).

D-6. $E = 0 - \frac{0.0591}{2} \log \frac{16}{4} = - \frac{0.0591}{2} \times 2 \log 2 = - 0.0591 \times 0.301 = - 0.0178 \text{ Volt.}$

If connected in reverse direction, $E = 0.0178 \text{ volt.}$



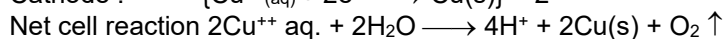
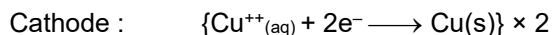
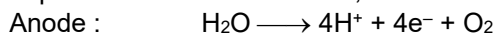
So conc. of Ag^+ will remain same.

E-2. equivalence of H_2 = equivalence of O_2

$$\frac{0.224}{22.4} \times 2 = \frac{\text{volume of } O_2}{22.4} \times 4$$

0.112 litre = volume of O_2 .

E-5. In presence of inert electrode, the cell reaction is -



Due to increases in $[H^+]$, pH decreases.

$\therefore$ (B) is answer



(1 mole)

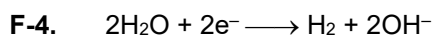
5 mole e^- = 5 Faraday.

F-2. Mole of Fe deposited = $\frac{1}{2} \times 3 = 1.5 \text{ mole}$

$$W_{Fe} = 1.5 \times 56 = 84 \text{ g.}$$

F-3. $W = \frac{63.5}{2 \times 96500} \times 2 \times 60 \times 60 = 2.37 \text{ g}$

$$\% \text{ of efficiency} = \frac{3}{2.37} \times 100.$$



$$\text{No. of Faraday passed} = \frac{9.65 \times 1000}{96500} = 0.1 \text{ F}$$

$$n_{\text{OH}^-} \text{ formed} = 0.1 \text{ mol}$$

$$n\text{NaOH} = 0.1 \text{ mol} \equiv \mathbf{4 \text{ g.}}$$

F-5. According to Faraday's second law

$$\frac{\text{mass of A}}{\text{equivalent mass of A}} = \frac{\text{mass of B}}{\text{equivalent mass of B}} = \frac{\text{mass of C}}{\text{equivalent mass of C}}$$

$$\frac{4.5}{15/n_1} = \frac{2.7}{27/n_2} = \frac{9.6}{48/n_3}$$

$$0.3n_1 = 0.1n_2 = 0.2n_3 = k$$

$$n_1 = \frac{10}{3}k$$

$$n_2 = 10k$$

$$n_3 = 5k$$

$$n_1 : n_2 : n_3 = \frac{10}{3} : 10 : 5 = \frac{1}{3} : 1 : \frac{1}{2} = 2 : 6 : 3$$

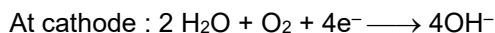
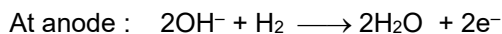
$$0.3 : 0.1 : 0.2$$

$$3 : 1 : 2$$

G-1. Discharging reaction



G-2. $\text{H}_2\text{-O}_2$ fuel cell



H-1. For strong electrolyte

$$\Lambda_M^C = \Lambda_M^\infty - b\sqrt{C}$$

H-2. Molar conductivity $\propto$ no. of ions per mole of electrolyte.

I-1. $K_a = 25 \times 10^{-6}$ $\wedge_{\text{eq}} = 19.6 \text{ Scm}^2 \text{ eq}^{-1}$, $C = 0.01$

$$K_a = 0.01 \times \alpha^2 \quad \Rightarrow \quad \alpha = \sqrt{\frac{25 \times 10^{-6}}{10^{-2}}} = 5 \times 10^{-2}$$

$$\alpha = 5 \times 10^{-2} = \frac{19.6}{\wedge_{\text{eq}}^\circ} \quad \Rightarrow \quad \wedge_{\text{eq}}^\circ = \frac{19.6}{5 \times 10^{-2}} = 392 \text{ Scm}^2 \text{ eq}^{-1}.$$

I-2. $1.53 = \frac{1000 \times 3.06 \times 10^{-6}}{\text{Normality}}$

$$\text{Normality} = 2 \times 10^{-3} \text{ M}$$

$$\text{Molarity} = \frac{2 \times 10^{-3}}{2} = 10^{-3} \text{ M}$$

$$K_{\text{sp}} = 10^{-6} \text{ M}$$

I-3. $K_a = C\alpha^2 = 0.1 \times \left(\frac{7}{380.8}\right)^2 = 3.38 \times 10^{-5}$





I-4. $K = 1.382 \times 10^{-6} \text{ s cm}^{-1}$

$$\Lambda_{\text{AgCl}} = 61.9 + 76.3 = 138.2 = \frac{1000 \times 1.382 \times 10^{-6}}{S} \Rightarrow S = 10^{-5} \text{ M.}$$

I-5. $\Lambda_{\text{m, BaSO}_4} = (x_1 + x_2 - 2x_3) \Rightarrow \Lambda_{\text{eq, BaSO}_4} = \frac{\Lambda_{\text{eq, BaSO}_4}}{n - \text{factor}}$

$$\Lambda_{\text{eq, BaSO}_4} = \frac{(x_1 + x_2 - 2x_3)}{2}$$

PART - III

1. (A) $E^\circ_{\text{Zn}^{2+}/\text{Zn}} > E^\circ_{\text{Mg}^{2+}/\text{Mg}}$ so $E^\circ_{\text{cell}} = -\text{ve}$
 (B) $E^\circ_{\text{Ag}^+/\text{Ag}} > E^\circ_{\text{Zn}^{2+}/\text{Zn}}$ so $E^\circ_{\text{cell}} = +\text{ve}$
 (C) $E^\circ_{\text{Ag}^+/\text{Ag}} > E^\circ_{\text{Zn}^{2+}/\text{Zn}}$ so $E^\circ_{\text{cell}} = 0$ (at equilibrium)
 (D) $E^\circ_{\text{Ag}^+/\text{Ag}} > E^\circ_{\text{Fe}^{2+}/\text{Fe}}$ so $E^\circ_{\text{cell}} = +\text{ve}$

2. (A) $E_{\text{cell}} = -\frac{0.059}{2} \log \frac{(P_{\text{H}_2})_c [\text{H}^+]_a^2}{(P_{\text{H}_2})_a [\text{H}^+]_c^2}$

$$E_{\text{cell}}^0 = -\frac{0.059}{2} \log \frac{0.01 \times (0.1)^2}{(0.1) \times (1)^2} = \frac{0.059}{2} \times 3 = +\text{ve}$$

(B) Cell reaction $\text{Ag}^+_c (10^{-2}) \rightarrow \text{Ag}^+_a (10^{-9})$

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.059}{1} \log \frac{10^{-9}}{10^{-2}} = E_{\text{cell}}^0 + \frac{0.059}{1} \times 7 > 0$$

$E_{\text{cell}}^0 \neq 0$ and not conc. cell

(C) $E_{\text{cell}} = 0 - \frac{0.059}{2} \log \frac{[\text{Cu}^{+2}]_a}{[\text{Cu}^{+2}]_c} = -\frac{0.059}{2} \log \frac{0.1}{0.01} = -\text{ve}$

(D) $E_{\text{cell}} = -\frac{0.059}{2} \log \frac{[\text{Cl}^-]_c}{[\text{Cl}^-]_a} = \frac{0.059}{2} \log \frac{0.1}{0.1} = 0$

And $E_{\text{cell}}^0 = 0$.

EXERCISE # 2

PART - I

1. As $E^\circ_{\text{Cu}^{2+}} \longrightarrow \text{Cu} = 0.337 \text{ V} > E^\circ_{\text{H}^+/\text{H}_2}$
 $\therefore \text{Cu}^{2+}$ can be reduced by H_2 .
2. Lower standard reduction potential related metal ions can displace higher standard reduction potential related metal ions.
3. (C) M is more reactive than carbon and B is more reactive than A. Also both B and A are less reactive than C.
4. Only anode or cathode can't work alone so absolute value of reduction potential can't be determined.



5. $E_1 = E^\circ - \frac{R}{n} \frac{T}{F} \ln 2$
 $E_2 = E^\circ - \frac{R}{n} \frac{2T}{F} \ln 1 = E^\circ$
 $\therefore E_2 > E_1$
6. $E_{\text{cell}} = 0.059 \log \frac{C_1}{C_2}$
 For E_{cell} to be +ve and maximum
 $\frac{C_1}{C_2} < 1$ or $C_1 < C_2$ Give $C_2 = 1\text{M}$.
 $\therefore C_1$ should be the minimum conc. of H^+ . $\therefore$ (B) is the right answer.
7. $E_{\text{cell}} = \frac{0 - 0.059}{1} \log \sqrt{\frac{10^{-5}}{10^{-3}}} = 0.059 \text{ V}$
8. $E_1 = -\frac{0.059}{1} \log [\text{H}^+]$
 or $\text{pH}_1 = E_1 / 0.059 = \text{pK}_a + \log \frac{x}{y}$
 $\text{pH}_2 = E_2 / 0.059 = \text{pK}_a + \log \frac{y}{x}$
 or $\frac{E_1 + E_2}{0.059} = 2 \text{pK}_a$ or $\text{pK}_a = \frac{E_1 + E_2}{0.118}$
9. $E_{\text{cell}} = (0.77 - 0.0713) - \frac{0.059}{1} \log \frac{0.02}{0.1 \times 0.34} = 0.713 \text{ volt.}$
10. Impure Cu will oxidise from anode along with Zn but only Cu^{2+} will reduce on cathode in purification of Cu^{2+} .
11. At Cathode : $2\text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{H}_2 + 2\text{OH}^-] \times 2$
 At Anode : $2\text{H}_2\text{O} \longrightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$
 $\text{H}_2 = 2 \text{ mole}$
 $\text{O}_2 = 1 \text{ mole}$ Total volume = $3 \times 22.4 = 67.2 \text{ L.}$
12. $\frac{1000 \times 2}{(55 + 32)} = \frac{27 \times 24 \times 3600 \times \eta}{96500}$ or $\eta = 0.951 = 95.1\%$
13. $\frac{9.72}{22.4} \times 2 = \frac{2.35}{22.4} \times 4 + \frac{W}{194} \times 2$ or $W = 43.47 \text{ g}$
14. After removing cathode no net charge will flow but ions move randomly.
15. Rusting reaction of Fe is -
 $2\text{H}^+ + \text{Fe} + \frac{1}{2} \text{O}_2 \longrightarrow \text{Fe}^{2+} + \text{H}_2\text{O}$



PART - II

1. (A) H_4XeO_6 has more SRP, so better oxidizing agent than F_2
 (B) O_3 has more SRP, so will oxidize Cl_2
 (C) $E^\circ_{\text{ClO}_4^-/\text{ClO}_3^-}$ is more in acidic medium than in basic medium
 (D) $E^\circ_{[\text{Fe}(\text{CN})_6]^{3-}/[\text{Fe}(\text{CN})_6]^{4-}} = 0.36 \text{ V}$
 Hence $[\text{Fe}(\text{CN})_6]^{4-}$ can be easily oxidized by ClO^- , Ce^{4+} , Br_2O^- but not by Li^+
 (E) $E^\circ_{\text{ClO}^-/\text{Cl}^-(\text{OH}^-)}$ is more than $E^\circ_{\text{BrO}^-/\text{Br}^-}$ and $E^\circ_{\text{ClO}_4^-/\text{ClO}_3^-(\text{OH}^-)}$ so true.
 (F) Ce^{4+} can't oxidize Cl_2 in acidic medium.

2. $\text{H}_2 \longrightarrow 2\text{H}^+ + 2\text{e}^-$
 1 atm 10^{-10}

$$E_{\text{H}_2/\text{H}^+} = 0 - \frac{0.059}{2} \log \frac{(10^{-10})^2}{1} \Rightarrow E_{\text{H}_2/\text{H}^+} = +0.59 \text{ V}$$

3. $\text{Cu}^{2+} + 4\text{NH}_3 \longrightarrow [\text{Cu}(\text{NH}_3)_4]^{2+}$
 $\begin{matrix} 1 & a & 0 \\ 0 & 2 & 1 \end{matrix}$

$$K_f = \frac{[\text{Cu}(\text{NH}_3)_4]^{2+}}{[\text{Cu}^{2+}] [\text{NH}_3]^4}$$

$$\text{or } 1 \times 10^{12} = \frac{1}{[\text{Cu}^{2+}] (2)^4} \text{ or } [\text{Cu}^{2+}] = 6.25 \times 10^{-14}.$$

$$\text{Now, } E_{\text{Cell}} = E_{\text{Cell}}^0 - \frac{0.059}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = 1.1 - \frac{0.059}{2} \log \frac{1}{6.25 \times 10^{-14}} = 0.71 \text{ V}.$$

4. $\lambda_m = \frac{K \times 1000}{M}$

$$200 = \frac{K \times 1000}{0.04}$$

$$K = \frac{200 \times 0.04}{1000} = 8 \times 10^{-3} \text{ S cm}^{-1}.$$

$$K = \left(\frac{\ell}{9} \right) \times \frac{1}{R}$$

$$8 \times 10^{-3} = \left[\frac{8}{4} \right] \times \frac{1}{R}$$

$$R = \frac{2}{8 \times 10^{-3}} = \frac{1}{4} \times 10^3 \Omega.$$

$$\text{Again } \Rightarrow V = IR$$

$$\text{So } I = \frac{V}{R} = \left(\frac{10}{\frac{1}{4} \times 10^3} \right) = 4 \times 10^{-2} \text{ A}.$$

5. $K_{\text{sol}} = K_{\text{Ba}^{2+}} + K_{\text{Ag}^+} + K_{\text{NO}_3^-}$

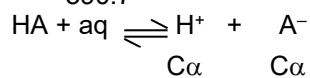
$$5.3 = \frac{\lambda_{\text{Ba}^{2+}}^\circ \times [\text{Ba}^{2+}]}{10^{-3}} + \frac{\lambda_{\text{Ag}^+}^\circ \times [\text{Ag}^+]}{10^{-3}} + \frac{\lambda_{\text{NO}_3^-}^\circ \times [\text{NO}_3^-]}{10^{-3}}$$



$$\therefore 5.3 = \frac{13 \times 10^{-3} \times 0.1}{10^{-3}} + \frac{6 \times 10^{-3} \times 0.2}{10^{-3}} + \frac{\lambda_{(\text{NO}_3^-)}^\circ \times 0.4}{10^{-3}}$$

$$\therefore \lambda_{(\text{NO}_3^-)}^\circ = 7 \times 10^{-3} \times 1000 \text{ Sm}^2\text{mol}^{-1} = 7 \text{ Sm}^2\text{mol}^{-1}$$

6. $\alpha = \frac{3.907}{390.7} = 0.1$



$$\text{C}\alpha - \text{C}\alpha$$

$$[\text{H}^+] = \text{C}\alpha = 0.01 \times 0.01 = 10^{-4}$$

$$\text{pH} = -\log [\text{H}^+] = 4$$

7. $\lambda^0 = \frac{1000 \times (\kappa_{\text{sol}} - \kappa_{\text{H}_2\text{O}})}{S}$

$$138 = \frac{1000 \times (3.4 - 2.02) \times 10^{-6}}{S}$$

$$S = 1 \times 10^{-5}$$

8. $\lambda_m^\circ (\text{H}_2\text{O}) = \frac{\lambda_m^\circ (\text{Ba}(\text{OH})_2) + 2\lambda_m^\circ (\text{HCl}) - \lambda_m^\circ (\text{BaCl}_2)}{2}$

$$= 5.5 \times 10^{-2} \text{ S m}^2 \text{ mol}^{-1}$$

Now, in SI units,

$$\lambda_m^\circ = \frac{K \times 10^{-3}}{M} \quad \therefore \lambda_m^\circ (\text{H}_2\text{O}) = \frac{K \times 10^{-3}}{[\text{H}_2\text{O}]_{\text{diss}}}$$

$$5.5 \times 10^{-2} = \frac{5.5 \times 10^{-6} \times 10^{-3}}{[\text{H}_2\text{O}]_{\text{diss}}}$$

$$[\text{H}_2\text{O}]_{\text{diss}} = [\text{H}^+] = [\text{OH}^-] = 10^{-7} \text{ M}$$

For equilibrium : $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$

$$\text{Dissociation constant } K = \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}_2\text{O}]} = \frac{10^{-7} \times 10^{-7}}{\left(\frac{1000}{18}\right)}$$

$$= 1.8 \times 10^{-16}$$

Reported answer = 18

9. H_3O^+ has highest Λ_m^∞ among cations. A doubly charged cation has higher Λ_m^∞ than unipositive cation.

PART - III

- Lower S.R.P. containing ion can displace higher S.R.P. containing ion.
- $\Delta G^\circ = -nFE^\circ_{\text{cell}}$
If $E^\circ_{\text{cell}} = +\text{ve}$ then $\Delta G^\circ = -\text{ve}$ and reaction is spontaneous.
- (A) Anode $2\text{H}_2\text{O} \longrightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$

(B) Anode $2\text{H}_2\text{O} \longrightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$

(C) Anode $\text{Cu} \longrightarrow \text{Cu}^{2+} + 2\text{e}^-$

(D) Anode $2\text{H}_2\text{O} \longrightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$



5. At Cathode : $\text{Cu}^{2+} + 2\text{e}^- \longrightarrow \text{Cu(s)}$
 At Anode : $\text{Cu (s)} \longrightarrow \text{Cu}^{2+} + 2\text{e}^-$

$$\text{Increase in mass of cathode} = \text{decrease in mass of Anode} = \frac{2.68 \times 3600}{96500} \times \frac{63.5}{2} = 3.174 \text{ g.}$$

6.

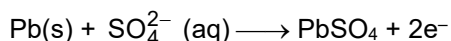
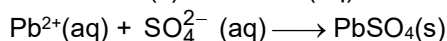
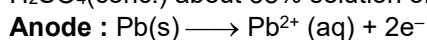
	Ag	Cu	Au
equivalent ratio	1	1	1
Mole ratio	$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{3}$
	6	3	2.

7. $\text{Fe}^{2+} + 2\text{e}^- \longrightarrow \text{Fe [in FeSO}_4\text{]} ; \quad \text{Fe}^{3+} + 3\text{e}^- \longrightarrow \text{Fe [in Fe}_2\text{(SO}_4\text{)}_3\text{]}$
 $\text{Fe}^{3+} + 3\text{e}^- \longrightarrow \text{Fe [in Fe (NO}_3\text{)}_3\text{]}$

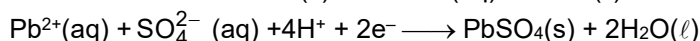
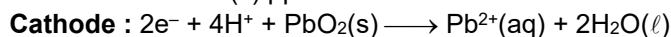
$$\text{Amount of Fe deposited in FeSO}_4 = \frac{Q}{96500} \times \frac{56}{2}$$

$$\text{Amount of Fe deposited in Fe}_2\text{(SO}_4\text{)}_3 = \frac{Q}{96500} \times \frac{56}{3}$$

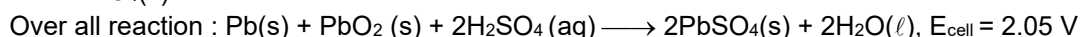
8. **Anode** : Pb(s)
Cathode : $\text{PbO}_2\text{(s)}$
 $\text{H}_2\text{SO}_4\text{(conc.)}$ about 38% solution of H_2SO_4 is taken.



most of the $\text{PbSO}_4\text{(s)}$ ppt sticks to the lead rod.



$\text{PbSO}_4\text{(s)}$ sticks to cathode rod.



9. Resistance of cell is not due to vibrations of ion but actually it is due to collisions of ions.

11. $\kappa \propto c$

Hence $\kappa_1 = a \times 0.1$

$\kappa_2 = a \times 0.01$

$\kappa_3 = a \times 0.005$

$\kappa_4 = a \times 0.0025$

$G = \frac{\kappa}{\text{cell constant}}$

So, $G_1 = \frac{a \times 0.1}{1} = 0.1 \text{ a} ; \quad G_2 = \frac{a \times 0.01}{10} = 0.001 \text{ a}$

$G_3 = \frac{a \times 0.005}{.5} = 0.001 \text{ a} ; \quad G_4 = \frac{a \times 0.0025}{25} = 0.0001 \text{ a}$

Hence $G_1 > G_2 = G_3 > G_4$

13. $\lambda_{\text{eq}}^\circ = \frac{\lambda_m^\circ}{V_f}$

V_f for cation/anion – charge; V_f for salt = total cationic/anionic charge

Also, $\lambda_m^\circ (\text{Al}_2(\text{SO}_4)_3) = 2 \lambda_m^\circ (\text{Al}^{3+}) + 3 \lambda_m^\circ (\text{SO}_4^{2-})$

& $\lambda_{\text{eq}}^\circ (\text{Al}_2(\text{SO}_4)_3) = \lambda_{\text{eq}}^\circ (\text{Al}^{3+}) + \lambda_{\text{eq}}^\circ (\text{SO}_4^{2-})$

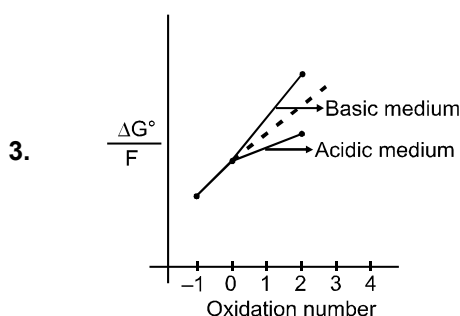


14. $\text{Li}^+ \text{Na}^+ \text{K}^+ \text{Rb}^+ \text{Cs}^+$
 degree of Hydration decreases
 Size of ions decreases
 ionic mobility increases
 $\text{H}^+_{(\text{aq})}$ has smallest size therefore show maximum mobility.

PART - IV

1. From given Latimer diagrams, $\text{Cl}_2 - \text{Cl}^-$ is independent of H^+ concentration.

2. $\Delta G^\circ = \Delta G_1^\circ + \Delta G_2^\circ$, using this $E^\circ = \frac{0.42 + 1.36}{2} \text{ V} = 0.89 \text{ V}$



4. As $\frac{\Delta G^\circ}{F}$ is low, stability is higher.

5. As $\frac{\Delta G^\circ}{F}$ is low, stability is higher so, +2 and 0 state is more stable than +1.

6. $\lambda_m^C = \lambda_m^\infty - b\sqrt{C}$

when $C_1 = 4 \times 10^{-4}$ $\lambda_m^C = 107$

and when $C_2 = 9 \times 10^{-4}$ $\Lambda_m = 97$

so $107 = \lambda_m^\infty - b \times 2 \times 10^{-2}$... (1)

$97 = \lambda_m^\infty - b \times 3 \times 10^{-2}$... (2)

$b = 1000$

$\Lambda_m = \lambda_m^\infty - b\sqrt{C} \Rightarrow \lambda_m^\infty = \Lambda_m + b\sqrt{C} = 107 + 10^3 \times 2 \times 10^{-2}$

$\lambda_m^\infty = 127 \text{ ohm}^{-1} \text{ cm}^2 \text{ mole}^{-1}$

7. For $25 \times 10^{-4} \text{ (M) NaCl}$ solution

$\Lambda_m = \lambda_m^\infty - b\sqrt{C} \Rightarrow \Lambda_m = 127 - 10^3 (25 \times 10^{-4})^{1/2}$

$\Lambda_m = 127 - 10^3 \times 5 \times 10^{-2} \Rightarrow \Lambda_m = 77$

But $\Lambda_m = \frac{K \times 1000}{M}$ $K = \left(\frac{\ell}{a}\right) \times \frac{1}{R}$

$\Lambda_m = \left(\frac{\ell}{a}\right) \times \frac{1}{R} \times \frac{1000}{M}$

$\Lambda_m = [\text{Cell constant}] \times \frac{1000}{R \times M}$

$\Rightarrow 77 = [\text{Cell constant}] \times \frac{1000}{1000 \times 25 \times 10^{-4}}$

Cell constant = $77 \times 25 \times 10^{-4} = 0.1925 \text{ cm}^{-1}$



8. For Na_2SO_4 solution

$$K = \left(\frac{\ell}{a} \right) \times \frac{1}{R} = \frac{0.1925}{400} = 4.81 \times 10^{-4} \text{ ohm}^{-1} \text{ cm}^{-1}$$

$$\Lambda_m = \frac{K \times 1000}{M} = \frac{4.81 \times 10^{-4} \times 1000}{\frac{5}{2} \times 10^{-3}}$$

$$\Lambda_m (\text{Na}_2\text{SO}_4) = 192.4 \text{ ohm}^{-1} \text{ cm}^2 \text{ mole}^{-1}$$

9. At the equivalence point the concentrations will be $[\text{Br}^-] = 100 \text{ mol/m}^3$, $[\text{Na}^+] = 100 \text{ mol/m}^3$
Therefore $\kappa_{\text{total}} = \kappa_{\text{Br}^-} + \kappa_{\text{Na}^+} = 1.2 \text{ Sm}^{-1} = 12 \times 10^{-1} \text{ Sm}^{-1}$.

EXERCISE # 3

PART - I

1. MnO_4^- ion can oxidise both Fe^{2+} to Fe^{3+} as well as Cl^- to Cl_2 . So $\text{Fe}(\text{NO}_3)_2$ cannot be estimated quantitatively with MnO_4^- ion in HCl .

E°_{Cell} for the cell $\text{Pt}, \text{Cl}_2(\text{g}) (1 \text{ atm}) | \text{Cl}^-(\text{aq}) || \text{MnO}_4^-(\text{aq}) | \text{Mn}^{2+}(\text{aq})$ is equal to $(1.51 - 1.4) = 0.11 \text{ V}$.

2. $E + 0.03 = E^\circ - \frac{0.06}{2} \log \frac{[\text{Zn}^{2+}]}{0.5}$.

$$E = E^\circ - \frac{0.06}{2} \log \frac{[\text{Zn}^{2+}]}{C}. \quad \mathbf{C = 0.05 \text{ M.}}$$

3. $\text{Zn} | \text{Zn}^{2+} (0.01 \text{ M}) || \text{Fe}^{2+} (0.001 \text{ M}) | \text{Fe}, \quad E = 0.2905$
cell reaction, $\text{Zn} + \text{Fe}^{2+} \rightleftharpoons \text{Zn}^{2+} + \text{Fe}$.

$$0.2905 = E^\circ - \frac{0.0591}{2} \log \frac{0.01}{0.001}$$

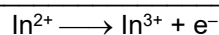
$$E^\circ = 0.32 \text{ Volt.}$$

At equilibrium, $E_{\text{cell}} = 0$.

$$0 = 0.32 - \frac{0.0591}{2} \log K_{\text{eq}}$$

$$K_{\text{eq}} = 10^{0.32/0.0295}$$

4. $\text{In}^+ \longrightarrow \text{In}^{3+} + 2\text{e}^-, \quad 0.42 \text{ volt.}$
 $\text{In}^{2+} + \text{e}^- \longrightarrow \text{In}^+, \quad -0.4 \text{ volt.}$

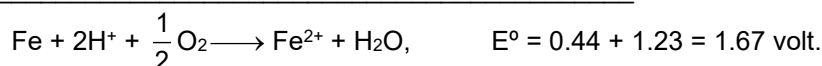


$$E^\circ = 0.44 \text{ volt.}$$

$$E^\circ_{\text{cell}} = 0.15 + 0.44 = 0.59 \text{ volt.}$$

$$0 = 0.59 - \frac{0.059}{1} \log K. \Rightarrow K = 10^{10}.$$

5. $\text{Fe} \longrightarrow \text{Fe}^{2+} + 2\text{e}^-, \quad 0.44 \text{ V.}$
 $2\text{H}^+ + \frac{1}{2} \text{O}_2 + 2\text{e}^- \longrightarrow \text{H}_2\text{O}, \quad 1.23 \text{ V.}$



$$\Delta G^\circ = -2 \times 1.67 \times 96500 = -322.3 \text{ kJ.}$$



6. $2\text{Ag}^+ + \text{C}_6\text{H}_{12}\text{O}_6 + \text{H}_2 \longrightarrow 2\text{Ag(s)} + \text{C}_6\text{H}_{12}\text{O}_7 + 2\text{H}^+$, $E^\circ = 0.8 - 0.05 = 0.75$ volt.
 $0 = 0.75 - \frac{0.0592}{2} \log K$.
 $\ln K = 2.303 \times \log K = 2.303 \times 25.34 = 58.38$.
7. $[\text{H}^+] = 10^{-11} \text{ M}$.
 $E_{\text{oxide}} = -0.05 - \frac{0.0591}{2} \log(10^{-11})^2 = -0.05 + 0.65$
 or, $\Delta H = 0.65$ volt.
8. Standards electrode potential does not depend upon on concentration.
9. $\text{AgBr(s)} \rightleftharpoons \text{Ag}^+ + \text{Br}^-$
 $(S + 10^{-7}) \times S = K_{\text{SP}} = 12 \times 10^{-14}$.
 $S = 3 \times 10^{-7} \text{ M}$.
 $[\text{Ag}^+] = 4 \times 10^{-7} \text{ M}$; $[\text{Br}^-] = 3 \times 10^{-7} \text{ M}$; $[\text{NO}_3^-] = 10^{-7} \text{ M}$.
 $K_{\text{total}} = \Lambda^\circ(\text{Ag}^+) \Lambda^\circ(\text{Ag}^+) + \Lambda^\circ(\text{Br}^-) \Lambda^\circ(\text{Br}^-) + \Lambda^\circ(\text{NO}_3^-) \Lambda^\circ(\text{NO}_3^-)$
 $\Lambda^\circ_{\text{KCl}} = \Lambda^\circ_{\text{K}^+} + \Lambda^\circ_{\text{Cl}^-}$.
 $K_{\text{KCl}} = 4 \times 10^{-4} \times 6 \times 10^{-3} + 3 \times 10^{-4} \times 8 \times 10^{-3} + 1 \times 10^{-4} \times 7 \times 10^{-3}$.
 $K_{\text{KCl}} = 24 + 24 + 7. \Rightarrow K_{\text{KCl}} = 55 \text{ Scm}^{-1}$.
10. Mol of NaCl = $4 \times 0.5 = 2$ mol.
 No. of mole of Cl_2 evolved = $\frac{1}{2} \times \text{mol of NaCl} = \frac{1}{2} \times 2 = 1$ mol.
11. Taking the 1 : 1 molar combination of Na–Hg amalgam.
 weight = $2 \times 23 + 2 \times 200 = 446$ g.
12. $2\text{Na}^+ + 2\text{e}^- \longrightarrow 2\text{Na}$.
 No. of Faraday required = 2.
 $\therefore$ total charge = $2 \times 96500 = 193000$ coulomb.
13. $\text{Cl}_2 + 2\text{I}^- \longrightarrow \text{I}_2 + 2\text{Cl}^-$.
 $E^\circ = 1.36 + (-0.54) = 0.82 \text{ V (+ve)}$. **Spontaneous.**
14. $\text{Mn}^{3+} + \text{e}^- \longrightarrow \text{Mn}^{2+}$, 1.50 volt.
 $2\text{H}_2\text{O} \longrightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$, 1.23 volt.
-
- $4\text{Mn}^{3+} + 2\text{H}_2\text{O} \longrightarrow 4\text{Mn}^{2+} + \text{O}_2 + 4\text{H}^+$, $E_{\text{cell}} = 1.5 - 1.23 = 0.27$ volt. (+ve).
 Mn^{3+} will oxidise H_2O .
15. Faraday law equivalents of H_2 produced = $\frac{I \times t \text{ (sec)}}{96500}$
 $0.01 \times 2 = \frac{10 \times 10^{-3} \times t}{96500} = 96500 \times 2 = t$
 $19.3 \times 10^4 \text{ sec} = t$
16. The species having less reduction potential with respect to NO_3^- ($E^\circ = 0.96 \text{ V}$) will be oxidised by NO_3^- .
 These species are V, Fe, Hg.
17. $\text{M(s)} | \text{M}^+(\text{aq}, 0.05 \text{ M}) || \text{M}^+(\text{aq}, 1 \text{ M}) | \text{M(s)}$
 Anode : $\text{M(s)} \longrightarrow \text{M}^+(\text{aq}) + \text{e}^-$
 Cathode : $\text{M}^+(\text{aq}) + \text{e}^- \longrightarrow \text{M(s)}$
-
- $\text{M}^+(\text{aq}) |_{\text{c}} \rightleftharpoons \text{M}^+(\text{aq}) |_{\text{a}}$



$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{1} \log \frac{M^{+}(\text{aq})|_a}{M^{+}(\text{aq})|_c}$$

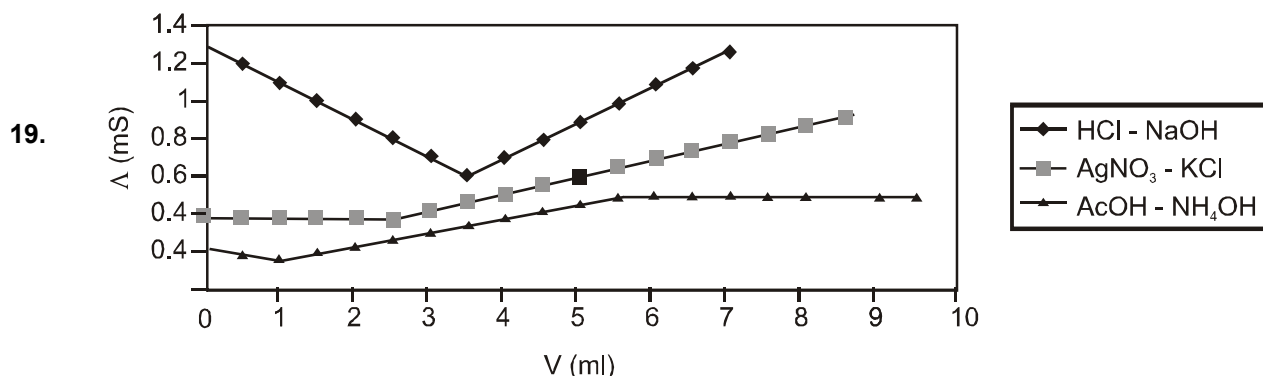
$$= 0 - \frac{0.0591}{1} \log \left\{ \frac{0.05}{1} \right\}$$

$$= +ve = 70 \text{ mV and hence } \Delta G = -nFE_{\text{cell}} = -ve.$$

18. $E_{\text{cell}} = \frac{-0.0591}{1} \log \left\{ \frac{0.0025}{1} \right\} = -\frac{0.0591}{1} \log \left\{ \frac{0.05}{20} \right\}$

$$= 70 \text{ mV} + \frac{0.0591}{1} \log 20 = 140 \text{ mV}.$$

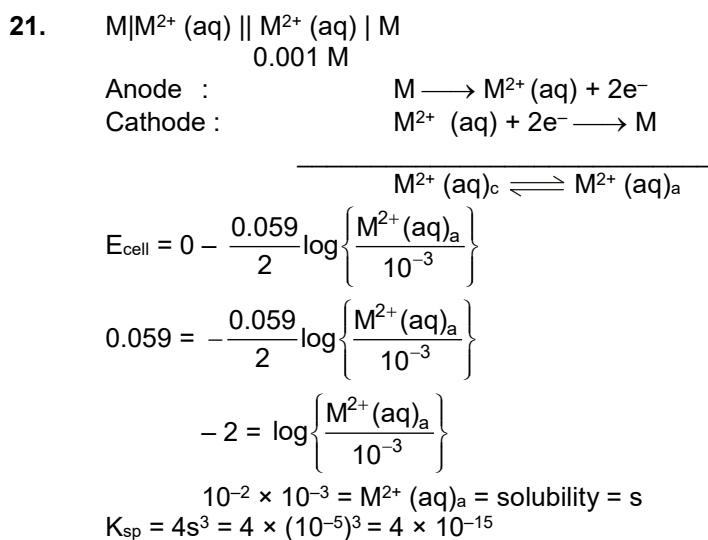
Typical titration curves



20. $E = E^{\circ} - \frac{0.059}{4} \log \frac{[\text{Fe}^{2+}]^2}{[\text{H}^{+}]^4 \text{P}_{\text{O}_2}}$

$$= 1.67 - \frac{0.06}{4} \log \frac{(10^{-3})^2}{(10^{-3})^4 \times 0.1} = 1.67 - \frac{0.03}{2} \log 10^7$$

$$= 1.67 - \frac{0.03}{2} \times 7 = 1.67 - 0.105 = 1.565 = 1.57 \text{ V}.$$



22. $\Delta G = -nFE_{\text{cell}} = -2 \times 96500 \times 0.059 \times 10^{-3} \text{ kJ/mole}$
 $= -11.4 \text{ kJ/mole}.$



26. $m^+ \longrightarrow m^{3+} + 2e^-$
 $\Delta G^0 = -nFE^0$ For 1 mole of m^+
 $\Delta G^0 = -2 \times 96500 \times (-0.25) \text{ J}$
 $= + 48250 \text{ J/mole} = 48.25 \text{ KJ/mole}$
 Energy released by conversion of 1 mole of
 $x \longrightarrow y \quad \Delta G = -193 \text{ KJ}$
 Hence mole of m^+ convert
 $\frac{193}{48.25} = 4$

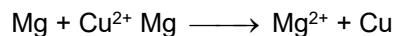
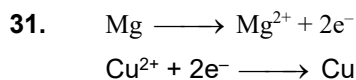
27. $\lambda_{X^-}^\circ \approx \lambda_{Y^-}^\circ$
 $\Rightarrow \lambda_{H^+}^\circ + \lambda_{X^-}^\circ \approx \lambda_{H^+}^\circ + \lambda_{Y^-}^\circ$
 $\Rightarrow \lambda_{HX}^\circ \approx \lambda_{HY}^\circ \quad (1)$
 Also $\frac{\lambda_m}{\lambda_m^\circ} = \alpha$, So $\lambda_m(HX) = \lambda_m^\circ \alpha_1$ and $\lambda_m(HY) = \lambda_m^\circ \alpha_2$
 (Where α_1 and α_2 are degrees of dissociation of HX and HY respectively.)
 Now, Given that
 $\lambda_m(HY) = 10 \lambda_m(HX)$.
 $\Rightarrow \lambda_m^\circ \alpha_2 = 10 \times \lambda_m^\circ \alpha_1$
 $\alpha_2 = 10 \alpha_1 \quad (2)$
 $K_a = \frac{C\alpha^2}{1-\alpha}$, but $\alpha \ll 1$, therefore $K_a = C\alpha^2$.
 $\Rightarrow \frac{K_a(HX)}{K_a(HY)} = \frac{0.01\alpha_1^2}{0.1\alpha_2^2} = \frac{0.01}{0.1} \times \left(\frac{1}{10}\right)^2 = \frac{1}{1000}$.
 $\Rightarrow \log(K_a(HX)) - \log(K_a(HY)) = -3. \quad \Rightarrow \quad pK_a(HX) - pK_a(HY) = 3.$

28. $E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.059}{2} \log_{10} \frac{[M^{2+}][H^+]^2}{[M^{4+}]pH_2}$
 $0.092 = 0.151 - \frac{0.059}{2} \log_{10} 10^x$
 $\therefore x = 2$

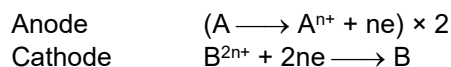
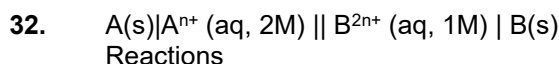
29. $C = 0.0015 \text{ M} \quad \ell = 120 \text{ cm}$
 $G = 5 \times 10^{-7} \text{ S} \quad a = 1 \text{ cm}^2$
 $G = \kappa \times \frac{a}{\ell}$
 $5 \times 10^{-7} = \kappa \times \frac{1}{120}$
 $\kappa = 6 \times 10^{-5} \text{ S cm}^{-1}$
 $\Lambda_m^c = \frac{\kappa \times 1000}{M} = \frac{6 \times 10^{-5} \times 1000}{0.0015}$
 $pH = 4$
 $[H^+] = 10^{-4} = c \alpha = 0.0015 \alpha$
 $\alpha = \frac{10^{-4}}{0.0015}$
 $\alpha = \frac{\Lambda_m^c}{\Lambda_m^\circ} \Rightarrow \frac{10^{-4}}{0.0015} = \frac{6 \times 10^{-5} \times 1000}{\Lambda_m^\circ}$
 $\Lambda_m^\circ = 6 \times 10^2 \text{ S cm}^2 \text{ mole}^{-1}$



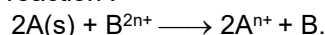
30. $\Delta G = \Delta G^\circ + 2.303 RT \log_{10} Q$; $Q = \frac{[Zn^{2+}]}{[Cu^{2+}]}$
 $= -2F(1.1) + 2.303 RT \log_{10} 10$
 $= 2.303 RT - 2.2 F$



$E = 2.67 = 2.7 - \frac{RT}{nF} \ln x$
 $0.03 = \frac{300}{2 \times 11500} \ln x$
 $2.3 = \ln x$
 $X = 10$



Overall reaction :



$E = E^\circ - \frac{RT}{2nF} \ln$

$O = E^\circ - \frac{RT}{2nF} \ln \frac{[A^{n+}]^2}{[B^{2n+}]}$

$E^\circ = \frac{RT}{2nF} \ln$

Now $\Delta G^\circ = -2nFE^\circ = \frac{-2nFRT}{2nF} \ln 4 = -RT \ln 4.$

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = 2\Delta G^\circ = -T\Delta S^\circ$

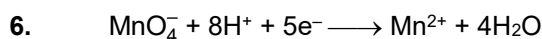
$T\Delta S^\circ = \Delta G^\circ$

$\Delta S^\circ = \frac{\Delta G^\circ}{T} = \frac{-RT \ln 4}{T} = -R \ln 4$

$= -8.3 \times 2 \times 0.7 = -11.62 \text{ JK}^{-1}\text{mol}^{-1}$

PART - II

5. Electrons flow from anode to cathode always.



$E = 1.51 - \frac{0.059}{5} \log \frac{[Mn^{2+}]}{[MnO_4^-][H^+]^8}$

Taking Mn^{2+} and MnO_4^- in standard state i.e. 1 M,

$E = 1.51 - \frac{0.059}{5} \times 8 \log \frac{1}{[H^+]} = 1.51 - \frac{0.059}{5} \times 8 \times 3 = 1.2268 \text{ V}$

Hence at this pH, MnO_4^- will oxidise only Br^- and I^- as SRP of Cl_2/Cl^- is 1.36 V which is greater than that for MnO_4^-/Mn^{2+} .

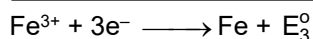
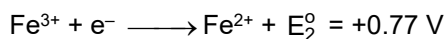
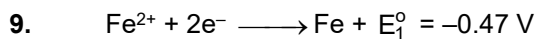


7. If a block of copper metal is dropped into a beaker containing solution of 1 M ZnSO_4 , no reaction will occur because $E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} = -0.76 \text{ V}$

$$E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} = +0.34 \text{ V}$$

Hence Cu can't displace Zn from ZnSO_4 solution.

8. Fact

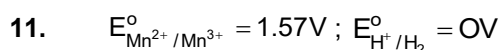


$$E_3^{\circ} = \frac{n_1 E_1^{\circ} + n_2 E_2^{\circ}}{n_1 + n_2} = \frac{2 \times (-0.47) + 1 \times (0.77)}{2 + 1} = -0.057 \text{ V}$$



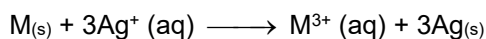
$$= E_{\text{A}^{\ell}}^{\circ} = +0.55 \text{ V}$$

Therefore $\text{T}^{\ell+}$ more stable



Mn^{2+} cannot reduce H^{+} to H_2

12. Cell reaction :



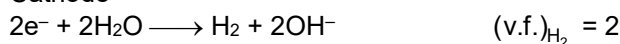
Applying Nernst equation :

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log_{10} Q$$

$$\therefore 0.421 = (0.8 \times E_{\text{M}^{3+}/\text{M}}^{\circ}) - \frac{0.059}{3} \log_{10} \frac{0.001}{(0.01)^3}$$

$$\therefore E_{\text{M}^{3+}/\text{M}}^{\circ} = 0.32 \text{ V}$$

13. Cathode

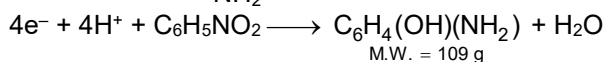
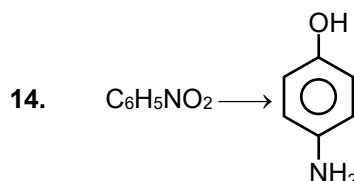


$$\text{mole} = \frac{i \times t}{\text{v.f.} \times 96500}$$

$$\frac{112}{22400} = \frac{i \times 965}{2 \times 96500}$$

$$\frac{1}{2} = \frac{i}{2}$$

$$i = 1 \text{ amp}$$



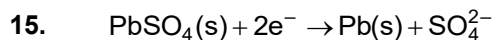
M.W. = 109 g



$$(v.f.) = 4 \quad W = ZIt = \frac{E}{F} \times I \times t \quad \left(E = \frac{M}{4} \right)$$

$$W = \frac{109 \times 9.65 \times 60 \times 60}{4 \times 96500}$$

$$W = 9.81 \text{ g}$$



For 2F current passed, PbSO_4 electrolyzed = 303g/mol

$$\text{For } 0.05\text{F}; \text{PbSO}_4 \text{ electrolyzed} = \frac{0.05 \times 303}{2} = 7.6\text{g}$$

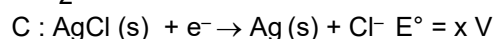
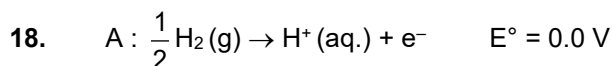
16. $\Delta G^\circ = -RT \ln K$

$$-2 \times 96000 \times 2 = -8 \times 300 \times \ln K$$

$$\text{or, } \ln K = 160$$

$$\text{or, } K = e^{160}$$

17. Higher the SOP, higher will be reducing power.



$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{1} \log \{[\text{H}^+][\text{Cl}^-]\}$$

$$0.92 = x - \frac{0.06}{1} \log (10^{-12})$$

$$0.92 = x + 0.72$$

$$x = 0.92 - 0.72 = 0.2 \text{ Volts}$$

19. Fact

20. $E^\circ_{\text{cell}} = E^\circ_{\text{Zn(s)}|\text{Zn}^{+2}} + E^\circ_{\text{Au}^{3+}/\text{Au}}$

$$= \text{SOP}_{\text{anode}} + \text{SRP}_{\text{cathode}}$$

$$= 0.76 \text{ V} + 1.40 \text{ V} = 2.16 \text{ V}$$

21. $E^\circ_{\text{cell}} = \frac{0.0591}{n} \log K_c = \frac{0.0591}{2} \log (1 \times 10^{16}) = 0.4728 \text{ V}$

22. $\Delta H = -nFE_{\text{cell}} + nFT \frac{dE}{dT}$

$$= -2 \times 96000 \times 2 + 2 \times 96000 \times 300 \times (-5 \times 10^{-4})$$

$$= -384000 - 28,800 = -412.8 \text{ kJ/mol}$$

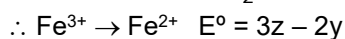
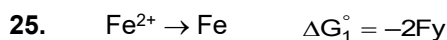
23. $\Lambda^\circ_{\text{m}}(\text{HA}) = \Lambda^\circ_{\text{m}}(\text{HCl}) + \Lambda^\circ_{\text{m}}(\text{NaA}) - \lambda^\circ(\text{NaCl})$

$$= 425.9 + 100.5 - 126.4 = 400$$

$$\Lambda^\circ_{\text{m}} = \frac{K \times 1000}{M} \Rightarrow \frac{5 \times 10^{-5} \times 10^3}{10^{-3}} = 50$$

$$\alpha = \frac{50}{400} = 0.125$$

24. Strongest oxidizing agent has highest value of SRP



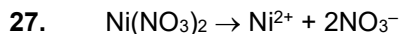
$$E_{\text{cell}}^\circ = E_{\text{Ag}^+|\text{Ag}}^\circ - E_{\text{Fe}^{3+}|\text{Fe}^{2+}}^\circ$$

$$= x - [3z - 2y] = x + 2y - 3z$$

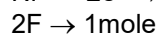
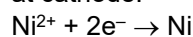
For given cell reaction

$$E_{\text{cell}}^\circ = x + 2y - 3z$$

26. $\Delta G^\circ = -nF E_{\text{cell}}^\circ = -2 \times 96000 \times 2.0 \times 10^{-3} = -384 \text{ kJ/mol}$

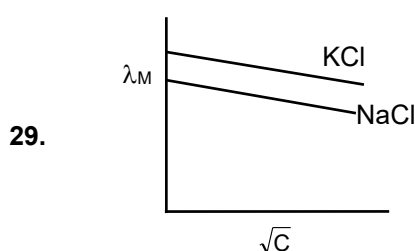


at cathode:



$$0.1F \rightarrow \frac{0.1}{2} = 0.05 \text{ mole}$$

28. Conductivity decreases on decreasing concentration of electrolyte.
Molar conductivity increases on decreasing concentration of electrolyte.

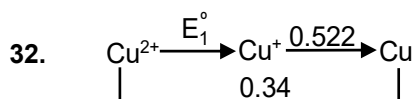


$$(\lambda_M^0)_{\text{K}^+} > (\lambda_M^0)_{\text{Na}^+}$$

Na^+ is more hydrated with respect to K^+ therefore KCl electrolyte have higher λ_M with respect to NaCl.

30. The metal ion with higher SRP value will have higher oxidising power.

31. Since acidic strength follows order $\text{HCOOH} > \text{C}_6\text{H}_5\text{COOH} > \text{CH}_3\text{COOH}$



$$2 \times 0.34 = E_1^\circ + 1 \times 0.522$$

$$E_1^\circ = 0.68 - 0.522 \Rightarrow E_1^\circ = 0.158 \text{ V}$$

33. Theory based.

34. From the given data

$$E_{\text{op}} = E_{\text{op}}^\circ - \frac{0.059}{4} \log [\text{H}^+]^4$$

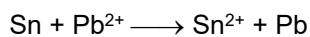
$$E_{\text{op}} = -1.23 - \frac{0.0591}{4} \log [\text{H}^+]^4$$

$$= -1.23 + 0.0591 \times \text{pH} = -1.23 + 0.0591 \times 5$$

$$= -1.23 + 0.2955 = -0.9345 \text{ V} = -0.93 \text{ V}$$



35. At Equilibrium state. $E_{\text{cell}} = 0$; $E^{\circ}_{\text{cell}} = 0.01 \text{ V}$

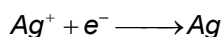


$$0 = 0.01 - \frac{0.06}{2} \log \left\{ \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]} \right\}$$

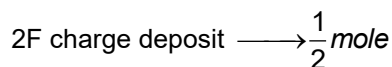
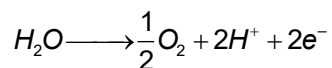
$$0.01 = \frac{0.06}{2} \log \left\{ \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]} \right\}$$

$$\frac{1}{3} = \log \left\{ \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]} \right\} \Rightarrow \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]} = 10^{1/3} = 2.1544$$

36. $(n_{\text{Ag}})_{\text{deposit}} = \frac{108}{108} - 1 \text{ mole}$



1F charge is required to deposit 1 mole of Ag



$$V_{\text{O}_2} = \frac{nRT}{P}$$

$$= \frac{1}{4} \times \frac{0.08314 \times 273}{1}$$

$$V_{\text{O}_2} = 5.674 \text{ L}$$

37. Theory based.